

Jiefu Zhang

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Research Interests

Reinforcement learning, robot navigation, multi-agent learning, inverse RL, LLM evaluation methodology, computer architecture, VLSI, ASIC

Education

Purdue University

Master of Science in Computer Engineering (Thesis Track)

Bachelor of Science in Computer Engineering

West Lafayette, IN

Expected December 2026

Graduated Dec 2024

- Accelerated BS/MS Program | Graduate GPA: 3.9/4.0
- **Relevant Coursework:** Artificial Intelligence, Embedded Systems, ASIC Design, VLSI Design, RISC-V Architecture, Semiconductor Fabrication

Publications & Manuscripts

- **J. Zhang** et al., "Don't Freeze, Don't Crash: Extending the Safe Operating Range of Neural Navigation in Dense Crowds." *arXiv:2603.06729*, 2026.
- Y. Xu, **J. Zhang**, H. Sun, Z. Zhou, T. Cao, V. Aggarwal, "Towards Reliable LLM Evaluation: Correcting the Winner's Curse in Adaptive Benchmarking." *arXiv:2605.05973*, 2026.
- H. Sun, Y. Xu, **J. Zhang**, X. Wu, Z. Zhou, J. He, J. Chen, "Core-Halo Decomposition: Decentralizing Large-Scale Fixed-Point Problems." *arXiv:2605.08681*, 2026.

Research Experience

Graduate Research Assistant

CLAN Lab, Purdue University

Advisor: Prof. Vaneet Aggarwal

Robot Navigation (PSS_Social)

Oct 2024 – Present

West Lafayette, IN

GitHub

- Developed a reinforcement learning framework for robot navigation in dense pedestrian crowds, introducing density-adaptive proxemic reward shaping and dynamic collision penalty tuning to address the freezing robot problem.
- Benchmarked against DS-RNN, SARL, ORCA, and SFM across random and circle-crossing scenarios at varying crowd densities, demonstrating improved success rates while reducing both freezing and collision failures.
- Designed and executed ablation studies to isolate contributions of individual reward components; analyzed performance under significantly denser conditions than prior benchmarks.
- First-authored paper available on [arXiv:2603.06729](#).

SIREN: Selection-Aware Inference for Adaptive LLM Evaluation

[arXiv:2605.05973](#)

- Co-developed SIREN, a selection-aware repeated-split reporting protocol that corrects the winner's curse in adaptive LLM benchmarking and delivers valid post-selection inference on the procedure-level performance of full tune-then-deploy pipelines under explicit tuning budgets.
- Designed a freeze-shortlist + splitwise selection / held-out evaluation scheme paired with an item-level Gaussian multiplier bootstrap, yielding simultaneous confidence bands on procedure-performance curves and supporting pre-specified equal-budget and cross-budget comparisons.
- Contributed to the theoretical analysis showing that, in a fixed-shortlist regime with smooth stabilized selection, the repeated-split estimator admits a first-order item-level representation that justifies bootstrap validity on a finite budget grid.
- Led MMLU-Pro tuning experiments across open-weight models (vLLM-served inference) with DSPy prompt optimization, demonstrating that winner-based reporting is optimistic and can flip deployment conclusions while SIREN remains close to the finite-sample target.

Core-Halo Decomposition for Decentralized Fixed-Point Problems

[arXiv:2605.08681](#)

- Co-developed the Core-Halo decomposition framework for decentralized stochastic approximation, separating disjoint update ownership (core) from overlapping read-only evaluation context (halo); proved exact fixed-point recovery under an explicit locality condition and a blockwise bias lower bound characterizing the structural failure of strict local decomposition.

- Built the SUMO multi-intersection traffic-control benchmark, including a 14-intersection Double-Corridor Bridge (DCB) topology and a 7-intersection barrier-ring topology, with PettingZoo-compatible Core-Halo environment wrappers and decentralized DQN training/evaluation pipeline; showed Core-Halo closes most of the gap to centralized DQN in mean waiting time using only one-hop read access.
- Implemented and verified the personalized PageRank experiments on synthetic stochastic block model graphs, empirically confirming the irreducible bias plateau of strict decomposition and the rapid near-centralized convergence of Core-Halo at contraction rate $\alpha = 0.85$.
- Contributed to the SARSA-based decentralized smart-grid energy management experiments on IEEE 9-, 14-, and 30-bus test systems, demonstrating Core-Halo recovers centralized-planner cost while strict decomposition incurs a strictly positive performance gap from truncated neighborhood-overload dependencies.

Inverse Reinforcement Learning for T2D Prescribing Behavior

- Developing an IRL framework to recover clinician decision logic from large-scale longitudinal pharmacy claims data, targeting Type 2 Diabetes drug subclass sequencing (metformin $\rightarrow$ sulfonylureas $\rightarrow$ GLP-1 agonists $\rightarrow$ SGLT2 inhibitors $\rightarrow$ insulin).
- Comparing MaxEnt IRL, Bayesian IRL, and Dynamic Discrete Choice formulations to model how provider specialty, geography, and practice patterns shape medication selection; downstream goal is offline RL for prescribing decision support.

Teaching Experience

Graduate Teaching Assistant
ECE 33700 - ASIC Design Laboratory

Jan 2025 – May 2025
West Lafayette, IN

- Instructing 60+ students in ASIC design; authored 10+ Verilog-based homework and lab assignments.

Selected Projects

Multicore RISC-V Processor

- Designed 5-stage pipelined processor supporting RISC-V ISA with hazard resolution via forwarding/stalling.
- Enhanced multicore efficiency by 30% using MOESI cache coherence protocol.

USB 1.1 ASIC Data Transfer Algorithm

- Achieved 12 Mbps throughput (USB 1.1 standard) using Verilog with AHB-Lite interface; validated via QuestaSim with 98% test coverage.

8-bit Wallace Tree Multiplier

ECE 559 MOS VLSI Design

- Designed and verified full schematic/layout with DRC/LVS-clean results; worst-case delay $\leq 2.5\text{ns}$, energy $\leq 850\text{fJ}$. Incorporated parasitic extraction for accurate timing and power evaluation.

Technical Skills

- **Programming:** Python, C/C++, Verilog, SystemVerilog, MATLAB, Java
- **ML/Research:** PyTorch, Deep RL, OpenAI Gym, PettingZoo, DSPy, vLLM
- **Hardware/EDA:** Cadence Virtuoso, QuestaSim, DRC/LVS, RTL Design, FPGA, ASIC/SOC Integration
- **General:** Linux, Git, $\LaTeX$

Additional

- **Languages:** English (fluent), Mandarin (native)
- **Industry Intern:** Product Maintenance Engineer Intern, Compound Co., Shenzhen (Summer 2023) – sensor calibration and IoT integration for medical devices.